

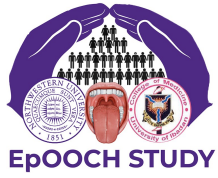
Epigenetic study of Oral HPV and Associated Oral/Oropharyngeal Cancer
in People living with HIV/AIDS

The role of Human Papilloma Virus (HPV) in Oropharyngeal Cancer (OPSCC)

Dr A. Daniel

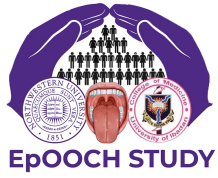
Oropharyngeal cancer





Introduction

- A sharp increase in the incidence of OPSCC has been observed in the last few decades.
- It presents in younger, fitter, more affluent patients who drink less alcohol and smoke less Tobacco
- It also presents with more aggressive disease with high prevalence of multiple cervical lymph nodes with extracapsular spread but paradoxically responds more favourably to treatment



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Anatomy of the Pharynx and Oropharynx

- Fibromuscular tube: 12 cm in the adult, extends from basiocciput to the sixth cervical vertebra

- Complete posteriorly, deficient anteriorly in three areas:

Nasopharynx

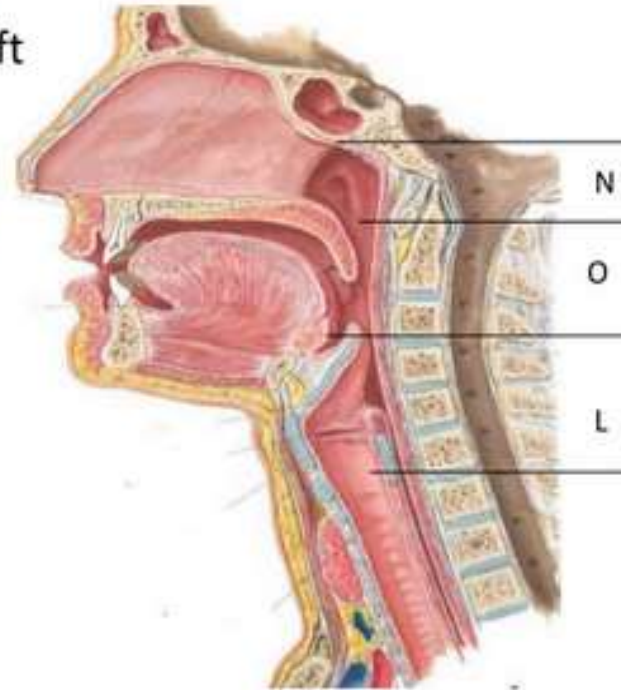
Oropharynx

Laryngopharynx/ Hypopharynx

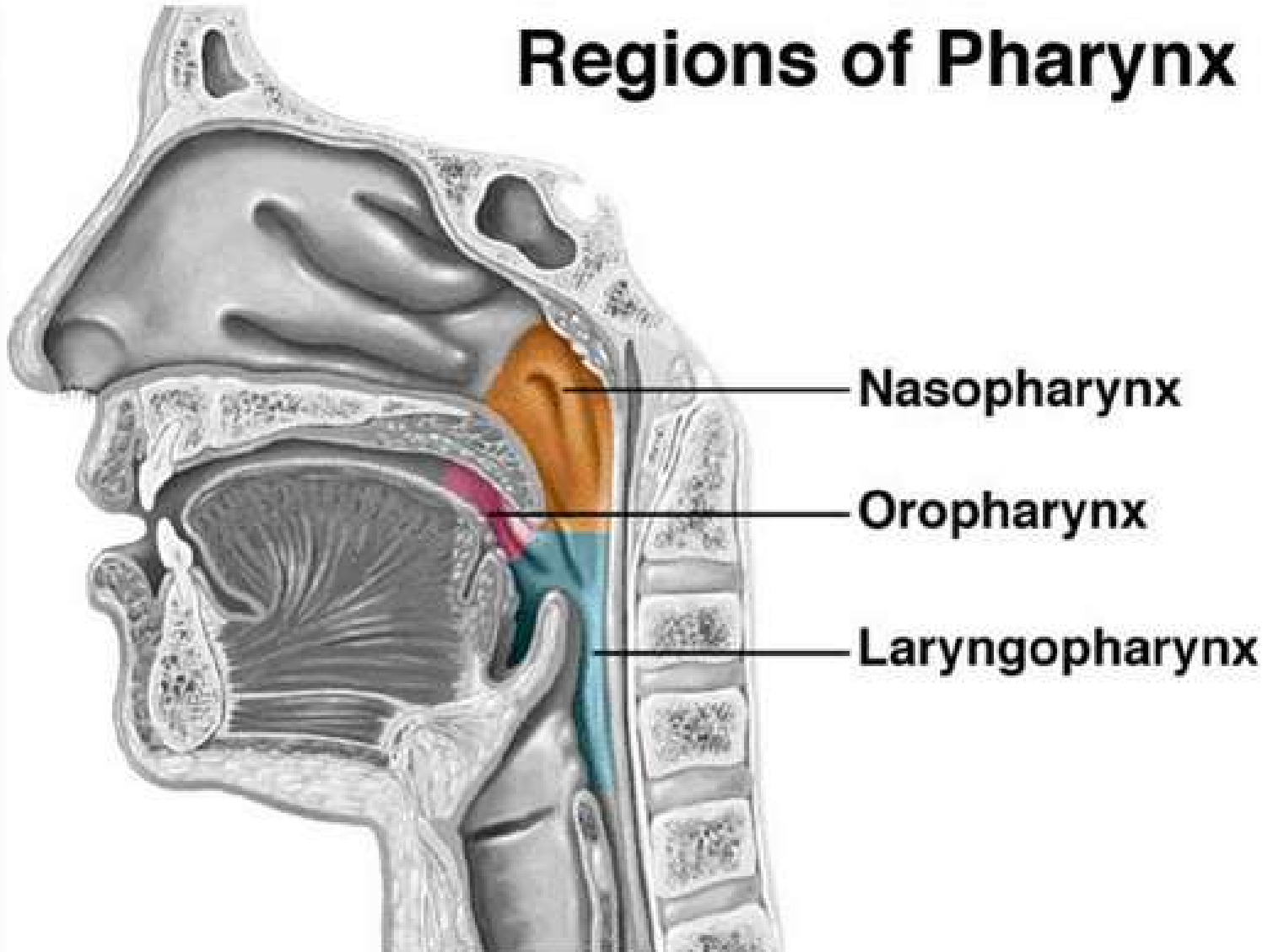
Anatomy of the Oropharynx

Division of Pharynx

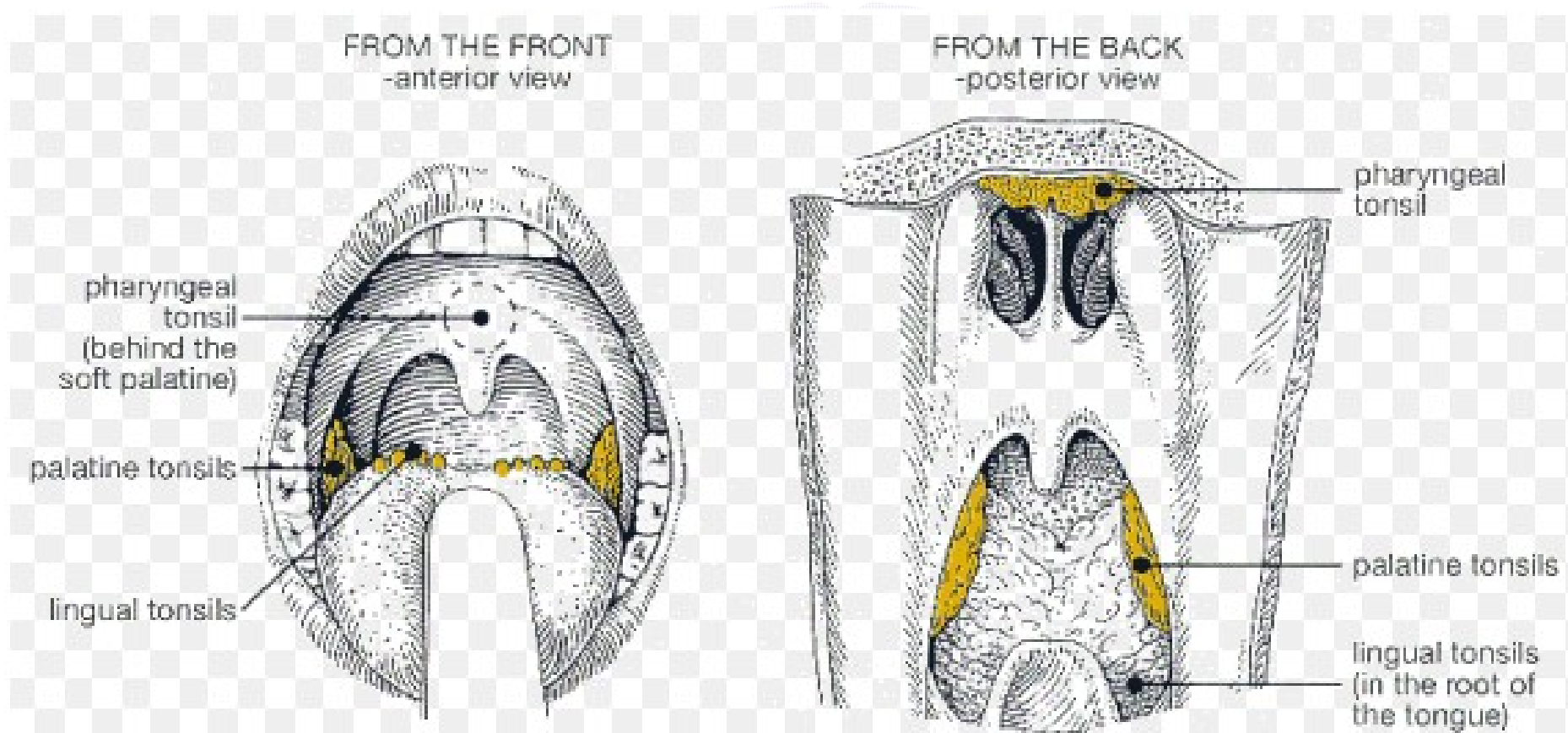
- The nasal part – **NASOPHARYNX/ EPIPHARYNX**
(extends from base of skull to soft palate)
- The oral part – **OROPHARYNX**
(extends from hard palate to hyoid bone)
- The laryngeal part – **LARYNGOPHARYNX/ HYPOPHARYNX**
(extends from upper border of epiglottis to lower border of cricoid cartilage)

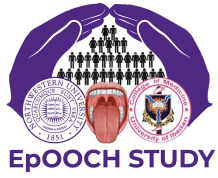


Regions of Pharynx



Oropharynx





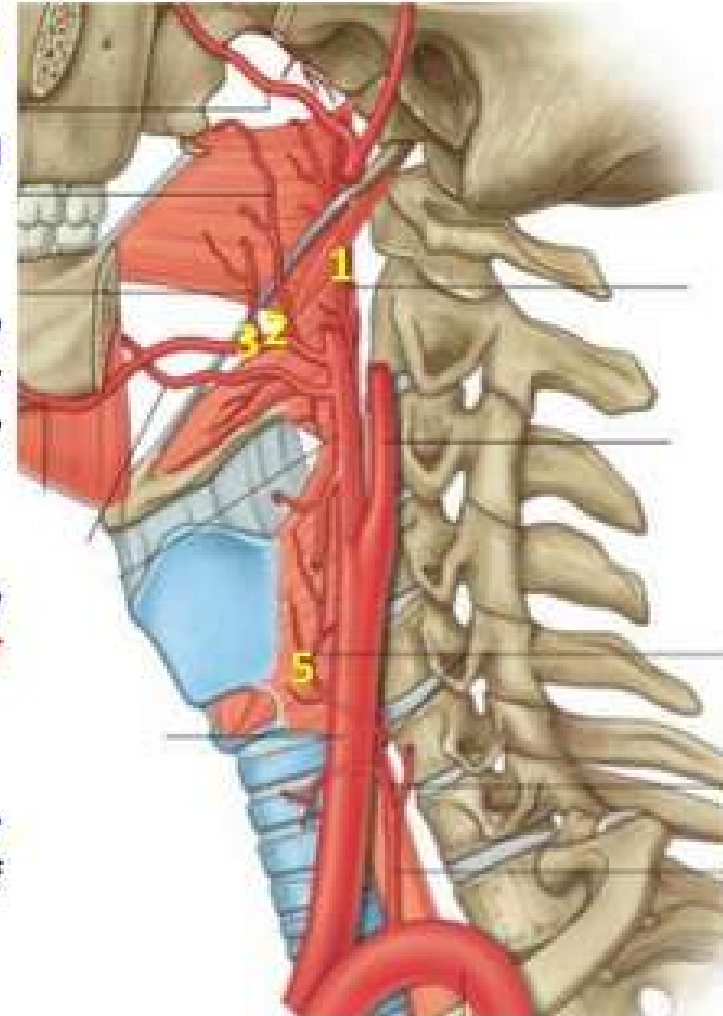
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Oropharynx

- Extends from the level of the soft palate cranially to the hyoid bone caudally
- Anterior wall is the tongue base extends to valleculae and lingual surface of the epiglottis
- Lateral wall: anterior and posterior faucial pillars and the palatine tonsil
- Posterior wall is the posterior pharyngeal wall from C2 to C3 vertebrae

Blood supply to pharynx

1. the ascending pharyngeal artery (ECA)
2. the ascending palatine branch of facial a. (ECA)
3. tonsillar branches of the facial artery (ECA) : penetrate superior constrictor muscle and it is the major blood supply to the tonsils
4. numerous branches of the maxillary and the lingual arteries. (ECA) **NOT SHOWN**
5. pharyngeal branches from the inferior thyroid artery (thyrocervical trunk of subclavian a.) :- supply howe pharynx



Venous drainage

PHARYNX

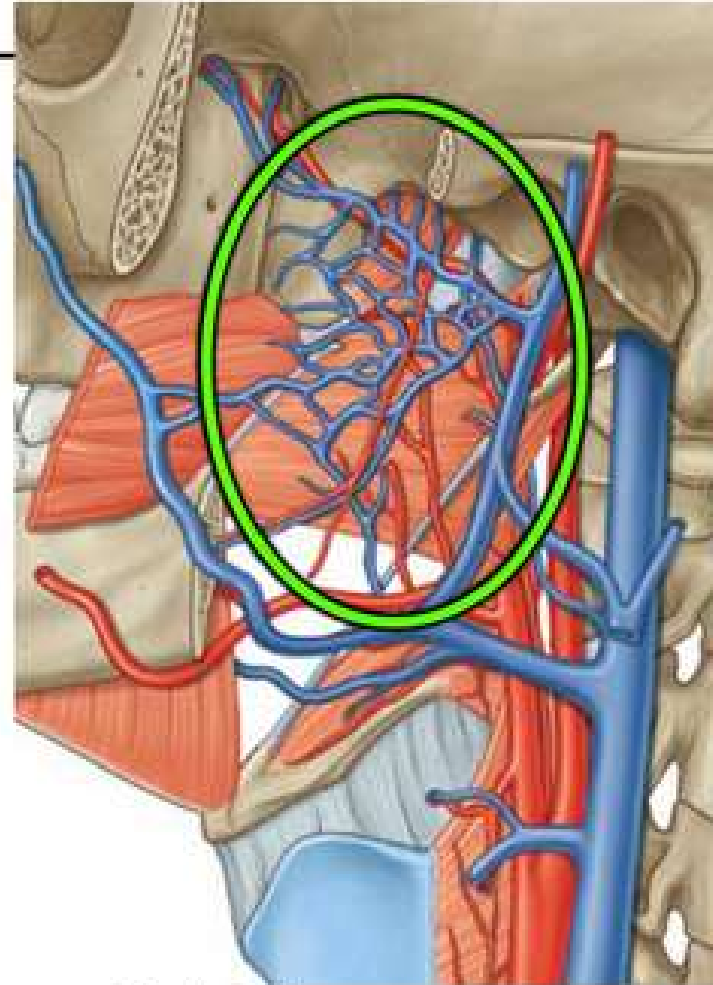
Venous

Venous

Mostly to **pterygoid venous plexus**
-including drainage from tonsillar
bed

Nasopharynx will also have drainage
along nasal vein paths

Laryngopharynx will also have drainage
along pathways to internal jugular vein



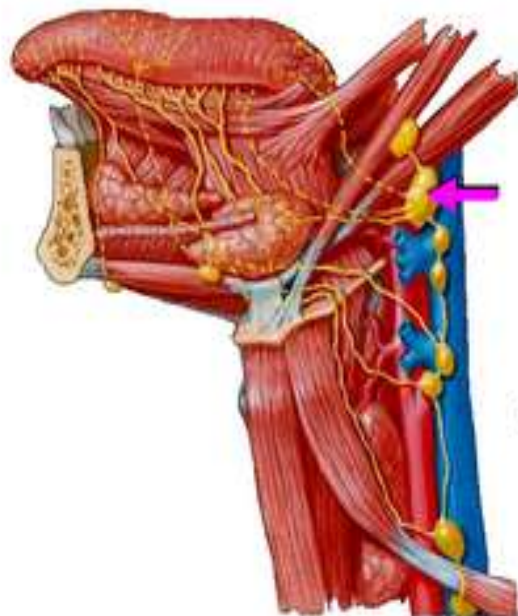
Gray's 947

Lymphatic drainage

PHARYNX

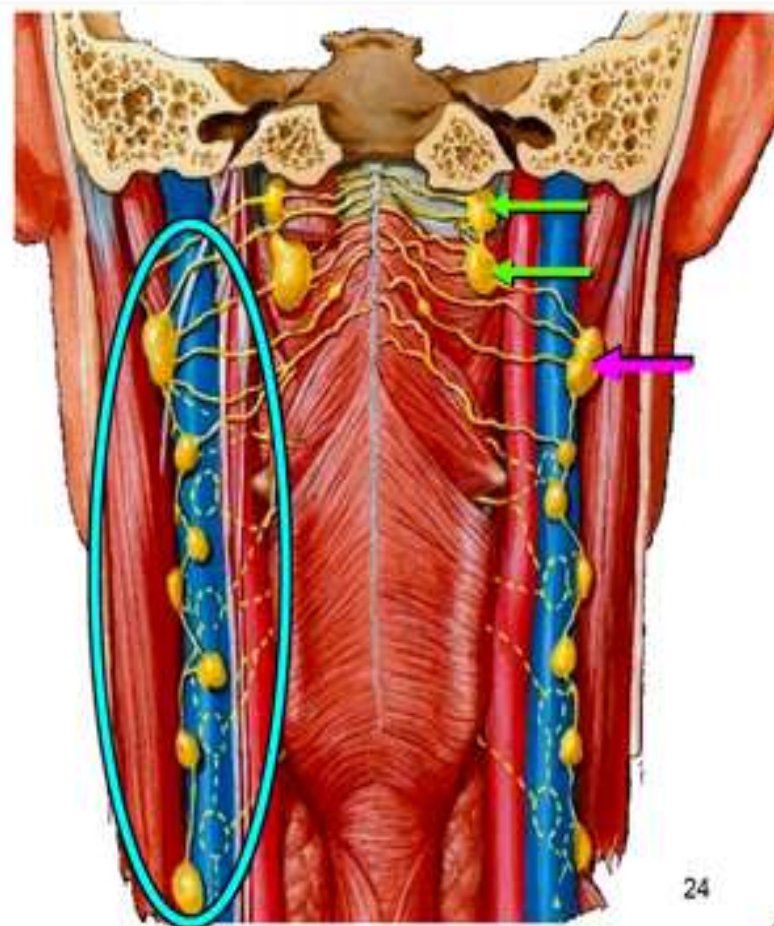
Most lymph drains to nodes near angle of mandible

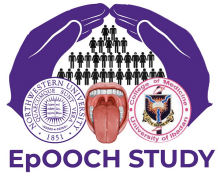
- some to **retropharyngeal nodes** ↑
- Most ends up in superior deep cervical nodes ○
- Superiormost is **jugulodigastric node** ↑



N 74, 75

Lymphatics

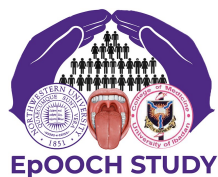




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Microscopic anatomy

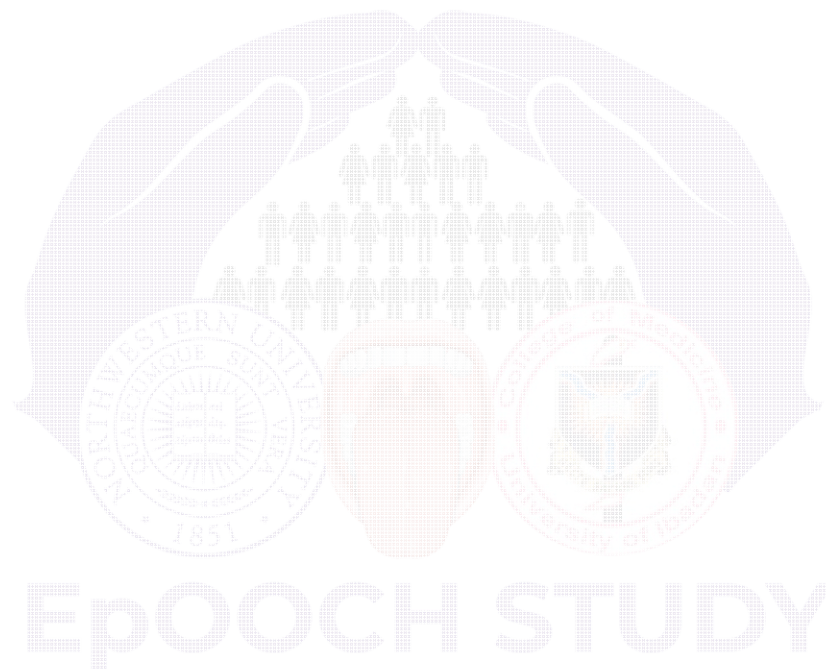
- Stratified squamous epithelium- islands of keratinization as we grow older
- Minor salivary glands
- Lymphoid tissue

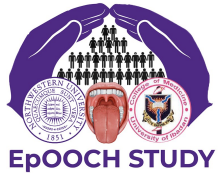


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Function of the oropharynx

- Swallowing
- Speech
- Defence





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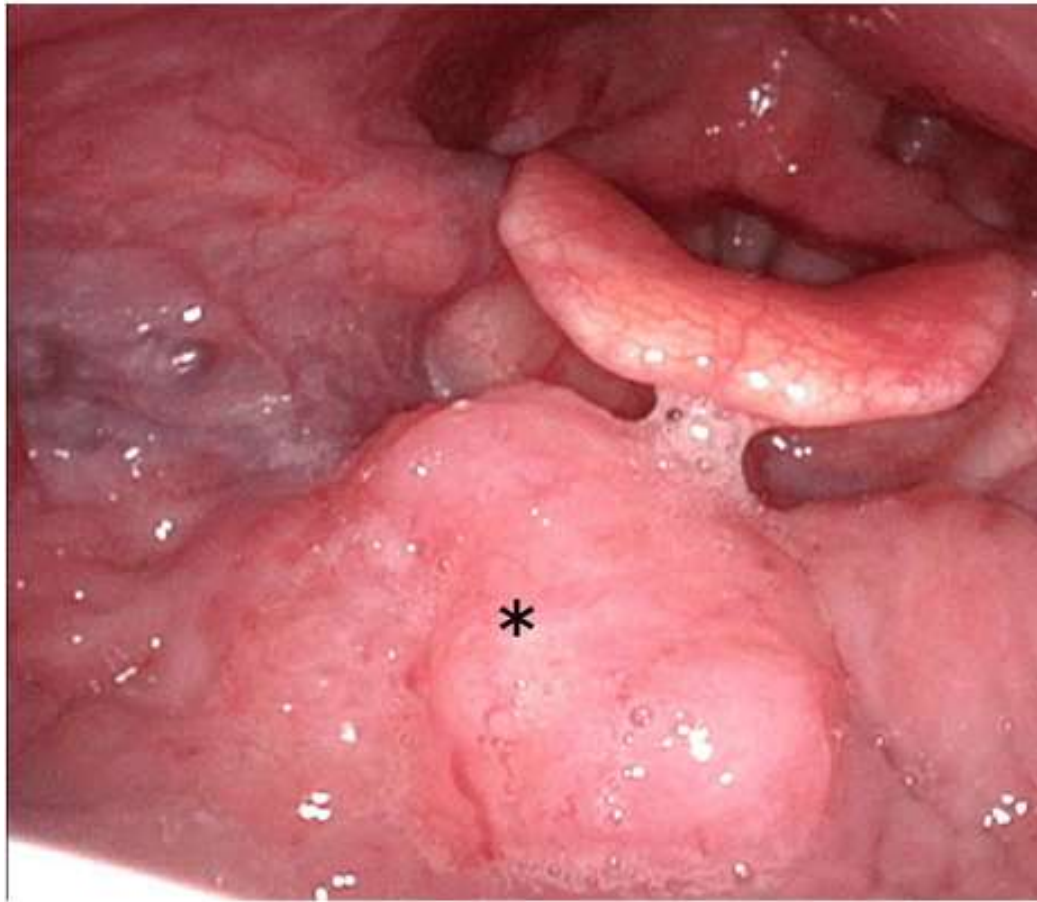
Oropharyngeal Malignancy

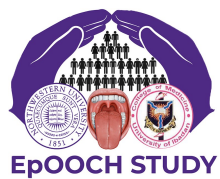
- Squamous Cell carcinoma is the commonest malignancy (90%)
- Lymphoma (Non-Hodgkin's Lymphoma)
- Minor Salivary Gland tumours
-
- Carcinoma of Unknown Primary*

Sites of Malignancy: lateral wall (60%)



Base of the Tongue (30%)



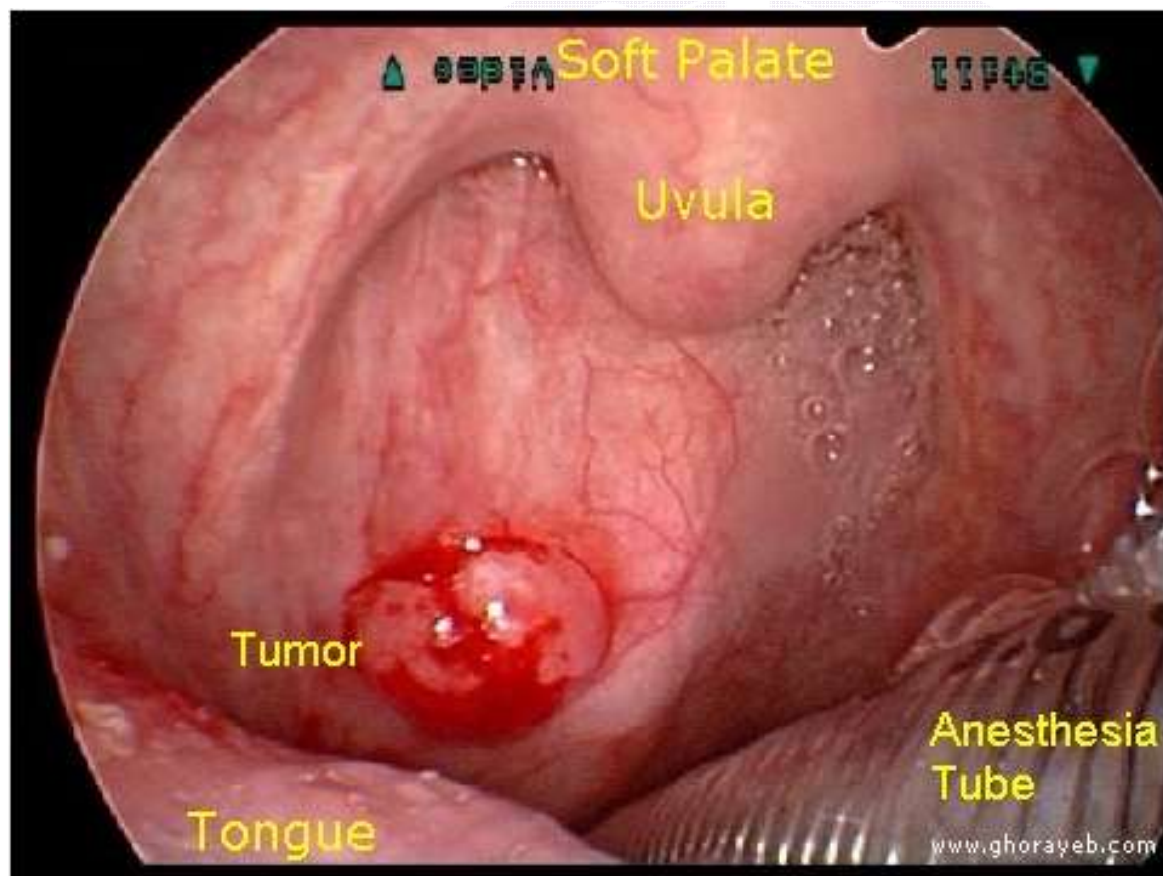


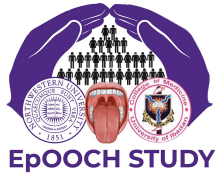
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Soft palate (5%)



Posterior Pharyngeal Wall (5%)





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Epidemiology

- **PREVIOUSLY:**

6-8/million in the United Kingdom (UK); 60/million in the United States of America (USA)

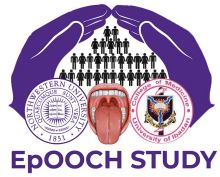
Age incidence 7th decade of life; M:F 10:1

Most significant risk factor: **Tobacco**

Alcohol and Tobacco- synergy

Tobacco & Lime

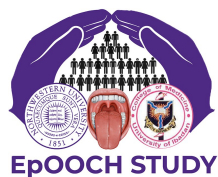
Ionizing radiation, iron deficiency anaemia, submucosal fibrosis of the palatal Arch, Dental sepsis



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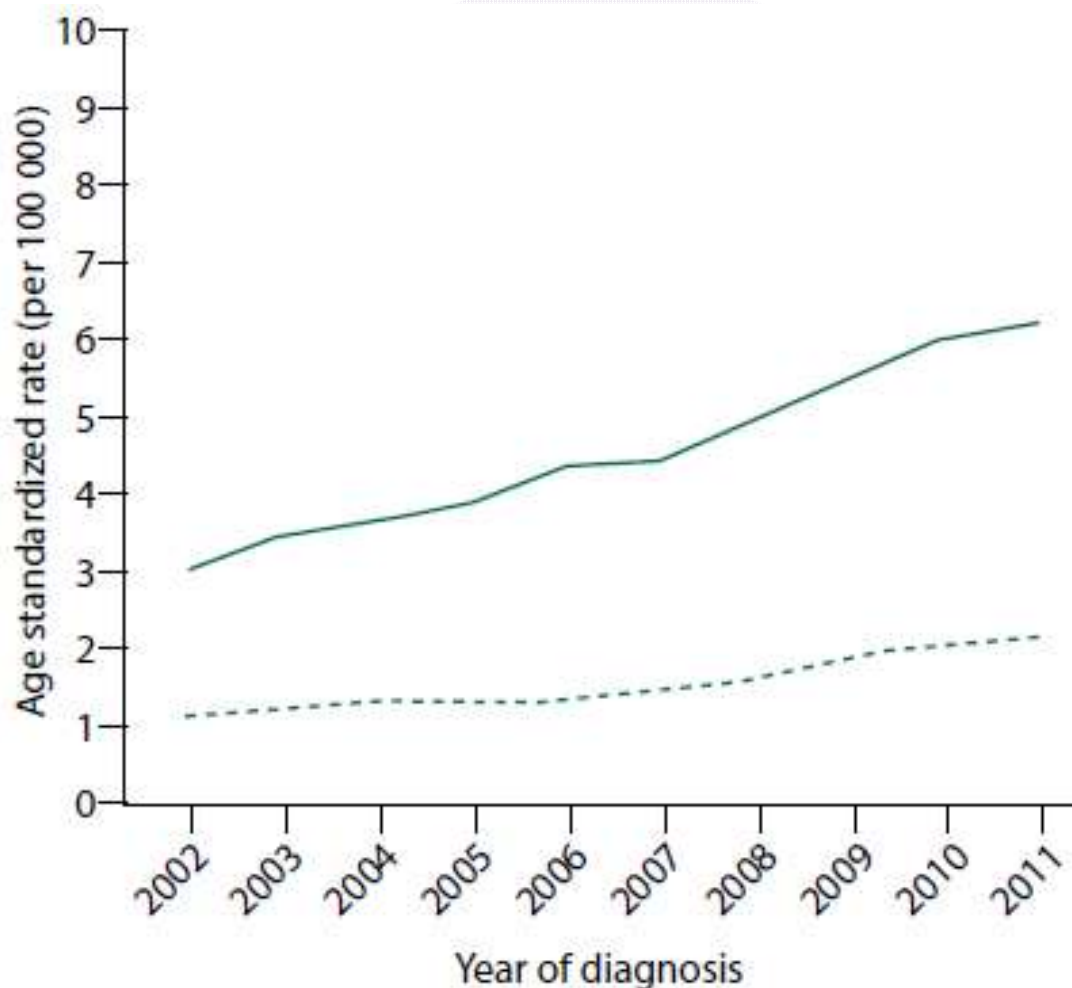
Recent trends

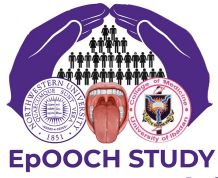
- Rapid increase in developed world (90% of OPSCC and 25% of OSCC are HPV related)
- 29,000 new cases of HPV related OPSCC estimated worldwide. Majority of cases in North America and Northern Europe
- UK: 2 fold increase between 2002 and 2011
- UK multicentre study of 1602 archival OPSCC histology blocks 51.8% demonstrated HPV 16
- HPV type 16 is an additional independent risk factor for OPSCC



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Age standardized rate of OPSCC in the UK 2002 - 2011

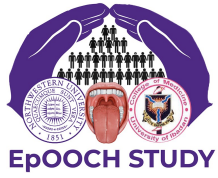




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- 4th to 5th decade of life
- M:F 4:1
- However, these figures are not consistent with figures seen worldwide between countries and time periods:

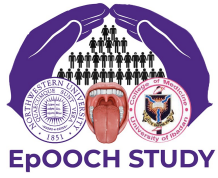
Substantial variation due to multiple factors including sexual behaviour, rates of genital HPV infection as well as tobacco and alcohol consumption



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Oral HPV infection

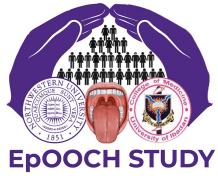
- Prevalence is much lower than genital infection
- Systemic review of 18 studies: 4500 healthy individuals oral infection with HPV- 4.5%, High Risk HPV (HRHPV)- 3.5%, HPV 16- 1.3%
- National Health and Nutrition Examination Survey (NHANE) 2012 in the USA – 5,500 healthy males and females:
- Oral HPV infection- 6.9%
- HRHPV- 3.9%
- HPV-16 – 1% (approx. 2 million people with oral HPV infection at any given point in time)



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Link between Oral HPV infection & OPSCC

- First postulated in 1938 by Sryjanen et al.
- Approximately 50% of oral rinses provided by patients with OPSCC had HPV-16
- OPSCC- 12 times more likely to have oral HPV than non-OPSCC (increases to 15fold with HPV-16)
- Patients with HPV+ve OPSCC have increased number of sexual partners
- Tobacco use also increased the risk of HPV+ve OPSCC (current and Past smokers)



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Molecular biology of HPV infection

- HPV: described in 2004 papillomaviridae, over 200 genotypes, non- enveloped double stranded DNA genome (12 genotypes are high risk)
- Obligate Epitheliotropic human pathogens
- Contains 8 genes: L1, L2, E1, E2, E4, E5, E6, E7
- E6 & E7 inhibit p53 and Retinoblastoma gene respectively



The diagram illustrates the pathways of cellular immortalization and malignant progression, showing the roles of various proteins and the effects of viral oncoproteins E6 and E7.

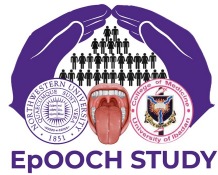
Key Components and Pathways:

- Cellular Cycle Regulation:** CDK2 and Cyclin E promote cell cycle progression. CDK4 and Cyclin D, regulated by P16, also contribute to cell cycle progression.
- Genomic Instability and Malignant Progression:** Genomic instability leads to malignant progression, which is further influenced by additional genetic alterations.
- Apoptosis Resistance:** P53 and E6AP regulate apoptosis resistance. E6 and E7 can lead to deregulation of DNA damage repair and cellular senescence.
- Cellular Immortalization:** hTERT promotes cellular immortalization, while telomere erosion inhibits it.
- Transcriptional Regulation:** NFY1 and E6AP regulate gene expression. E7 and RB regulate the CUL2 complex.
- Epigenetic Regulation:** ABL, RB, HDAC, DP-1, and E2F-1 regulate gene expression.

Diagram Labels:

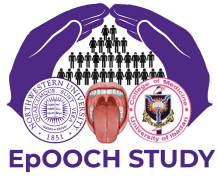
- A:** RB, E7, CUL2 complex
- B:** E7, P27, Cyclin E
- C:** E7, P21
- D:** P53, E6, E6AP, P21, Apoptosis resistance, Genomic instability, Malignant progression
- E:** P16, CDK4, CDK6, Cyclin D
- F:** NFY1, E6, E6AP, hTERT, Telomere erosion
- P:** ABL, RB, HDAC, DP-1, E2F-1

Cellular Compartments: Cytoplasm, Nucleus



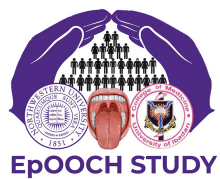
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- HPV 16 & 18 are the most important causing over 70% of HPV related cancers (oropharynx, cervix, vaginal, vulva, anal, penile)
- HPV -16 greater dominance in OPSCC compared to cervical cancer (95% vs 60%)
- Perhaps different mucosal environments may support specific genotypes



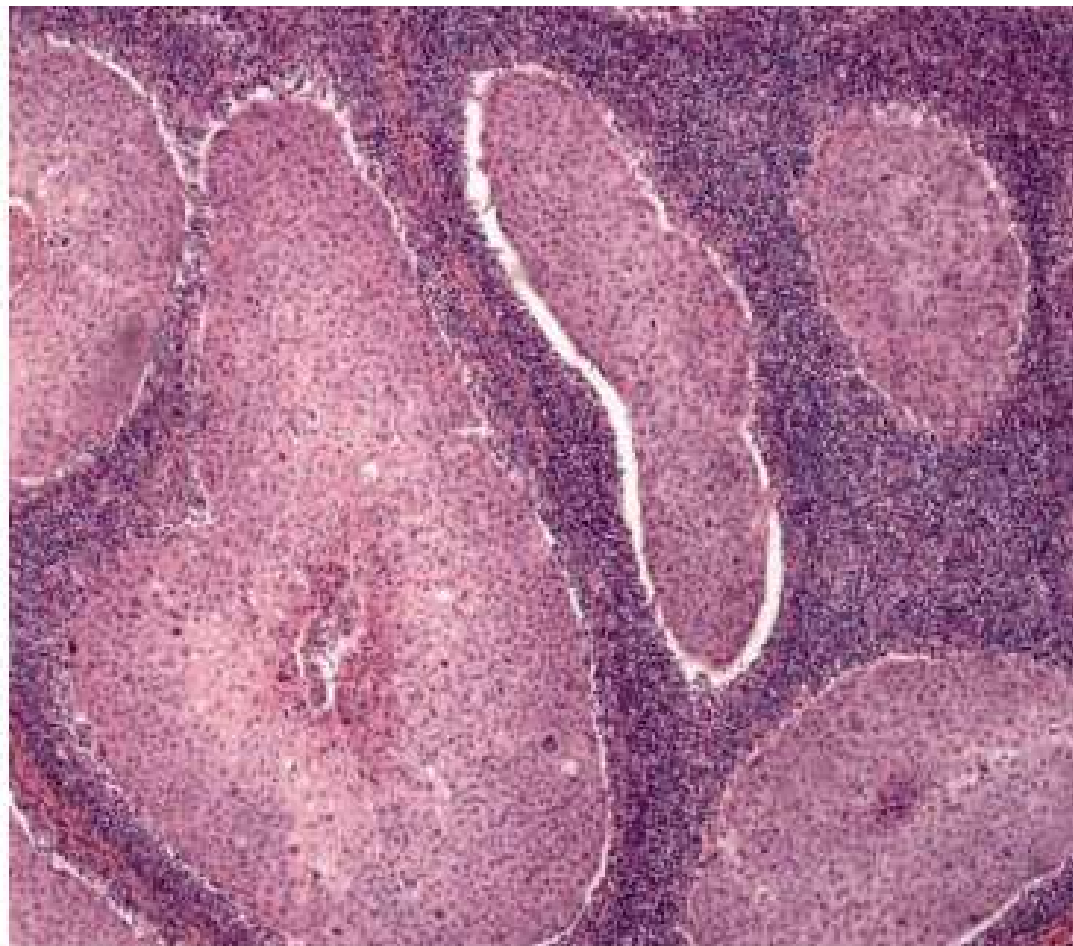
Histopathology

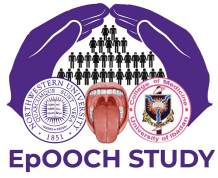
- HPV negative SCC- typically moderately differentiated and keratinizing tumours arising from dysplastic epithelium
- HPV+ve SCC: non-keratinizing, display poorly differentiated basaloid tumours not associated with surface dysplasia but associated with high level of Tumour Infiltrating Lymphocytes (TILs), the density of which have prognostic value



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H&E stain of OPSCC with TILs





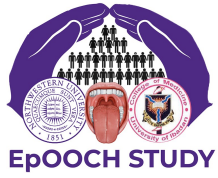
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Tests for HPV DNA

- In-situ Hybridization (ISH)
- Polymerase Chain Reaction (PCR)

Tests to determine biological activity

- Quantitative Reverse Transcription –Polymerase Chain Reaction (qRT-PCR)
- RNA based Chromogenic in-situ-hybridization (RNA-ISH)



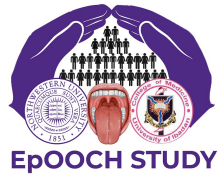
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Clinical features

- HPV+ve OPSCC: small primary tumours with multiple enlarged cystic cervical lymph nodes with high prevalence of extra capsular spread
- HPV-ve OPSCC: larger tumours, lower burden of cervical lymph nodes.

Treatment

- Same treatment: Surgery, radiation therapy, chemotherapy depending on the stage of the disease.
- HPV+ve OPSCC better prognosis



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Thank you for your attention

